MODEL TEAM / MATH 582

Building a linear dual SVM from the textbook

A technical explanation of the code, mathematics, dataset preparation, numerical optimizer, and test results.

What this delivers

The program **epilepsy_linear_dual_svm.py** implements the soft-margin **dual support vector machine** from Chapter 12, Section 12.3 of *Mathematics for Machine Learning*. It builds the quadratic objective and constraints itself, solves for the Lagrange multipliers, recovers a linear decision boundary, and predicts a binary seizure outcome.

> **Scope boundary:** This is only the Model Team deliverable: a basic linear dual SVM with a fixed soft-margin penalty **C = 1**. It does **not** implement nonlinear kernels, kernel comparisons, or hyperparameter searches.

Data and label definitions

Training sheet	5,000 synthetic records
Testing sheet	5,000 different synthetic records
Predictors	26 survey fields; 33 numerical columns after basic category encoding
Target	**Had a Seizure Two Weeks After Survey**
Output labels	**1 = seizure** within the simulated two-week follow-up; **0 = no seizure**

The dataset is entirely synthetic. These results are useful for demonstrating the mathematics and checking software, not for real-world clinical seizure prediction.

1 | From labels to the dual optimization

The dataset uses labels 0 and 1. The SVM equations in the book use labels −1 and +1. Convert them **internally** using

$$y_i^{SVM} = 2\, y_i^{data} - 1 \text{ (therefore } 0 \to -1,\ 1 \to +1)$$

Start with the soft-margin primal model (book, Section 12.2):

$$\min_{w,b,\xi} \tfrac{1}{2} ||w||^2 + C \Sigma_i \xi_i$$

subject to $y_i(w^T x_i + b) \geq 1 - \xi_i$ and $\xi_i \geq 0$

Introduce Lagrange multipliers α_i for the margin constraints. Taking derivatives of the Lagrangian with respect to w and b gives the identities below (book, Section 12.3):

$$w = \Sigma_i \alpha_i y_i x_i$$

$$\Sigma_i \alpha_i y_i = 0$$

The resulting convex dual quadratic program is Equation (12.41):

$$\min_{\alpha} \tfrac{1}{2} \Sigma_{i,j} \alpha_i \alpha_j y_i y_j (x_i^T x_j) - \Sigma_i \alpha_i$$

subject to $0 \leq \alpha_i \leq C$ for every i; $\Sigma_i \alpha_i y_i = 0$

In matrix notation, define $Q_{ij} = y_i y_j (x_i^T x_j)$. Then the code minimizes $\tfrac{1}{2}\boldsymbol{\alpha}^T Q \boldsymbol{\alpha} - \mathbf{1}^T \boldsymbol{\alpha}$. The dot product is the ordinary linear inner product; this implementation has **no nonlinear kernel**.

Why this is convex: the matrix Q is positive semidefinite, since it is a label-weighted Gram matrix. Minimizing this quadratic subject to linear and box constraints is a convex optimization problem.

2 | How the code solves the dual

A direct generic optimizer becomes expensive for 5,000 training points because there are 5,000 dual variables and a 5,000 × 5,000 Gram matrix. The code therefore uses an iterative **two-coordinate working-set quadratic optimizer**, which solves the *same* dual program.

Numerical procedure

- Build the linear Gram matrix $K = XX^T$, where each entry is $x_i^T x_j$.
- Initialize every $\alpha_i = 0$. This already satisfies the box limits and the equality constraint.
- At each step, select two multipliers whose joint adjustment can reduce the quadratic objective.
- Compute the exact one-dimensional quadratic-minimum step, then clip it to the allowable interval $0 \le \alpha_i, \alpha_j \le C$.
- Update the pair while preserving $\Sigma \alpha_i y_i = 0$; repeat until the Karush–Kuhn–Tucker (KKT) violation is sufficiently small.

Feasible pair update

$\alpha_i \leftarrow \alpha_i + d$; $\alpha_j \leftarrow \alpha_j - (y_i y_j)d$

The two updates cancel under the equality constraint. The bounds are enforced when choosing d.

Recovering the actual classifier

After optimization, the code computes $\mathbf{w = \Sigma\, \alpha_i y_i x_i}$ (book Eq. 12.38). For training examples satisfying $0 < \alpha_i < C$, it computes $\mathbf{b = y_i - w^T x_i}$ (book Eq. 12.42), then averages those estimates.

Score: $f(x) = w^T x + b$ | Predicted label: 1 if $f(x) \ge 0$; otherwise 0.

Fixed settings and convergence check

Soft-margin penalty C	1.0 (fixed baseline)
KKT stopping threshold	0.003
Actual KKT gap	0.002890
Equality residual $\lvert\Sigma\, \alpha_i y_i\rvert$	7.11e-15
Iterations / support vectors	76,402 / 2,892

The solver reached the specified KKT threshold. This establishes approximate satisfaction of the optimization conditions, not a guarantee that all predictions are correct.

3 | Dataset handling and held-out evaluation

The spreadsheet has a **Training_Data** sheet and a separate **Testing_Data** sheet. The former is used to fit the model; the latter is reserved for the final evaluation.

Minimal preprocessing

- Remove **Record ID** and the outcome from the inputs; neither is a predictor.
- One-hot encode **Sex** (2 categories) and **Type of Epilepsy** (7 categories). This is basic numerical representation, not a nonlinear SVM kernel.
- Standardize the other 24 numerical/coded survey inputs using the training means and standard deviations.
- Apply those same training-derived transformations to the 5,000 test records without refitting or using test labels.

This creates a 33-dimensional input vector: 24 standardized features plus 2 + 7 indicator columns. Survey fields represented as ordered numeric codes remain a simplifying baseline assumption; those integers are *not* validated clinical distances.

Held-out confusion matrix: 5,000 synthetic test records

Actual / predicted	Predicted 0	Predicted 1
Actual 0 — no seizure	2,176 (TN)	556 (FP)
Actual 1 — seizure	677 (FN)	1,591 (TP)

Accuracy	75.34%	Correct predictions among all test cases
Seizure recall / sensitivity	70.15%	Fraction of actual seizure cases identified
No-seizure specificity	79.65%	Fraction of actual non-seizure cases identified
Seizure precision	74.10%	Fraction of predicted seizures that were actual seizures

Interpretation: The model correctly classified 3,767 of 5,000 simulated records, but missed 677 of 2,268 simulated seizure events. This is an educational baseline and must not be used for clinical decisions.

4 | Running and verifying the model

Files to keep together

epilepsy_linear_dual_svm.py	The model, preprocessing, constrained dual optimizer, and classifier.
Training_Data.csv	5,000 original training rows converted from the spreadsheet.
Testing_Data.csv	5,000 original test rows converted from the spreadsheet.

Execute from that folder in a terminal

```
pip install numpy
python epilepsy_linear_dual_svm.py
```

The script prints model diagnostics and held-out metrics. It also saves **linear_svm_results.json** and **linear_svm_model.npz** (weights, intercept, training scaling information). The optional command below is only for a quicker mathematical demonstration; it does not give the same full-training result.

```
python epilepsy_linear_dual_svm.py --train-rows 250
```

Where the mathematics appears in the code

- **read_csv()** loads the numeric predictors and labels.
- **BasicPreprocessor** fits training-only numerical encoding and scaling.
- **solve_dual_linear()** constructs $K = XX^T$, iteratively optimizes α subject to the exact constraints, then computes w and b.
- **measure()** computes accuracy and the seizure/no-seizure confusion counts.

Numerical cross-check

On a 60-row training subset, the working-set optimizer was independently checked against the general SciPy SLSQP constrained minimizer. Both converged, with an absolute difference of about $\mathbf{7.4 \times 10^{-11}}$ in their minimized dual objectives. This verifies the small-case implementation against a general quadratic optimizer; it does not establish clinical validity.

References

Deisenroth, M. P., Faisal, A. A., & Ong, C. S. *Mathematics for Machine Learning*, Chapter 12, especially Sections 12.2–12.3 and Eqs. (12.38), (12.41), (12.42). https://mml-book.github.io/

MATH 582, *SVM Mini Project*, supplied project instructions, dated October 29 / due November 10. Dataset workbook: *Epilepsy_SVM_Synthetic_Dataset.xlsx*, supplied Codebook and ReadMe sheets.